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Research Assignment 1

Section A : Data Fundamentals

1. What is a Database?

A data base is an organized collection of data that is stored electronically so it can be easily accessed, managed, updated and analyzed. Data bases are important in the modern world because businesses, schools, banks, hospitals and social media platforms all rely on data to operate efficiently.

Databases help organizations:

- Store large amounts of information safely
- Retrieve information quickly
- Improve decision-making
- Reduce errors and duplication

Real world examples:

1. Banks use data bases to store customer accounts, balances and transactions.
2. Hospitals use databases to manage patients' records, appointments and medical histories.

2. What is a Data Warehouse?

A data warehouse is a system designed to store large amounts of historical data from multiple sources for reporting and analysis.

Difference between a database and a data warehouse:

- A regular database is mainly used for daily operations such as processing transactions.
- A data warehouse is used for analytics, reporting and business intelligence.

Companies use data warehouses to analyze trends, create reports, support business decisions and/or combine data from different systems.

A retail company, for example, may use a data warehouse to analyze sales from all branches over several years.

3. What is a data Lake?

A data lake is a storage system that holds raw structured, semi-structured and unstructured data.

The difference between a data lake and a data warehouse is that data warehouses store cleaned and organized data whereas data lakes store raw data in its original format.

Businesses prefer data lakes when:

- Handling very large volumes of data.
- Working with videos, images or social media content.
- Using machine learning and big data analytics.

An example would be a streaming company storing video viewing data and customer activity.

4 What is a Data Mart?

A data mart is a smaller section of a data warehouse focused on a specific department or business area. It relates to a data warehouse because it usually receives data from the main warehouse.

Typical uses of a Data Mart:

- Sales teams
- Finance departments
- Marketing departments.

For example, A sales data mart may only contain customer purchases and sales performance data.

5. Structured, Semi-Structured and Unstructured Data.

Structured Data

Structured data is organised into rows and columns.
Eg. Customer records stored in a SQL database.

Semi-Structured Data

Semi-structured data does not follow a strict table format but still contains labels or tags.

Eg. JSON or XML files.

Unstructured Data

Unstructured data has no pre defined format.

Eg. Videos, images, emails and social media posts.

6. What is Data Modeling?

Data modeling is the process of designing how data will be stored, organised and related within a system.

Importance of data modeling:

- Improves database design
- Reduces data duplication
- Makes systems efficient
- Helps developers understand relationships between data.

Types of data models:

1. Conceptual Data model - Gives high-level overview of the system.
2. Logical Data model - Shows relationships, attributes and rules.
3. Physical Data Model - Describes how the database will actually be implemented.

7. What is Data Mining?

Data mining is the process of discovering patterns, trends and useful information from large data sets.

Businesses use data mining to predict customer behaviour, detect fraud, improve marketing strategies and increase profits.

An example in the real-world would be banks using data mining to detect suspicious transactions or online stores using data mining to recommend

products to customers

8. What is a DBMS?

A Database Management System (DBMS) is a software used to create, manage and control databases.

MySQL

A popular relational database used for websites and applications.

PostgreSQL

An advanced open-source relational database known for reliability. Used in Financial systems and analytics.

Oracle Database

A powerful enterprise database used by large organizations. Used in banking and enterprise applications.

Mongo DB

A NoSQL database that stores flexible document-based data. Used in social media and big data applications.

9. What is ETL?

ETL stands for Extract, Transform, Load.

Extract → Data is collected from different sources

Transform → Data is cleaned, formatted and organised.

Load → The transformed data is then loaded into a database or data warehouse.

ETL is important because it helps organisations combine data from many systems into one clean and usable format for analysis.

10. Common Data Storage Formats

CSV

Stores data in plain text separated by commas.
Best for simple spreadsheets and data sharing.

JSON

Stores data in key-value format. Best for APIs and web applications.

Parquet

A compressed column based format optimized for analytics. Best for big data and fast querying.

Avro

A compact binary format often used in big data systems. Best for data streaming and distributed systems.

Section B: Artificial Intelligence & Machine Learning.

11 What is Artificial Intelligence?

Artificial Intelligence (AI) is the ability of computers or machines to perform tasks that normally require human intelligence.

Narrow AI (Weak AI)

AI designed for a specific task. Eg. Siri or Google assistant.

General AI (Strong AI)

AI that can perform any intellectual task like a human. True general AI does not fully exist yet.

12 What is Machine Learning?

Machine Learning (ML) is a branch of AI where computers learn patterns from data instead of being directly programmed.

The difference from traditional programming is that traditional programming follows fixed rules and Machine learning learns from examples and improves over time.

Types of ML:

1. Supervised Learning - Uses labeled data
2. Unsupervised Learning - Finds patterns in unlabeled data.
3. Reinforcement Learning - Learns through rewards and penalties.

13. What is Deep Learning?

Deep learning is a subset of machine learning that uses artificial neural networks with many layers.

Relationship:

- AI is the broad field.
- Machine learning is a subset of AI
- Deep learning is a subset of machine learning.

Neural networks are systems inspired by the human brain. They process information through connected layers of nodes to recognise patterns.

14. What is NLP?

Natural Language Processing (NLP) is a field of AI that allows computers to understand and process human language.

We find this in real-world applications such as chat bots like ChatGPT, Google Translate and in voice assistants like Siri and Alexa.

15. What is a Large Language Model (LLM)?

A Large Language Model is an AI model trained on massive amounts of text data to understand and generate human language.

Difference from standard ML models:

- Standard ML models are often designed for specific tasks.
- LLMs can perform many language tasks such as answering, writing and summarising.

Examples:

1. Chatgpt - used for conversations, writing and coding help.
2. Claude - used for research, writing and analysis.

16. What is Generative AI?

It is AI that creates new content such as text, images, music or videos. It works by learning patterns from existing data and generating similar content.

We find examples in ChatGPT that generates text responses, Midjourney which generates AI images and GitHub Copilot that assists developers with code generation.

17. What is a Training Dataset?

A training dataset is the collection of data used to teach an AI model. Quality is important here because poor-quality data can produce inaccurate or unfair results.

Problems caused by biased data are incorrect predictions, discrimination or unfair outcomes. For example, a biased AI (hiring AI) may unfairly reject certain groups of people.

18. What is Computer Vision?

Computer vision is a field of AI that enables machines to interpret and understand images and videos. Machines analyze patterns, colours, shapes and objects in visual data.

Cases where it is used would be; facial recognition systems, self-driving cars detecting roads and pedestrians.

Section C: APIs, Tools and Developer Concepts

19. What is an API?

An API (Application Programming Interface) is a system that allows software applications to communicate with each other.

Take an API as a waiter in a restaurant, the waiter takes your order to the kitchen and brings the food back. APIs are important because they connect applications, share data between systems, and automate processes.

20. What is an API key?

An API key is a secret code used to identify and authenticate a user or application. Its purpose is to control access to APIs, track usage, and improve security.

API Keys should be kept private, stored securely and never shared publicly. If exposed publicly, attackers may misuse the API and cause financial or security problems.

21. What is a Rest API?

REST stands for Representational State Transfer. It is a common style of API communication over the internet HTTP methods.

- GET → Retrieve data
- POST → Create data
- PUT → Update data
- DELETE - Remove data

22. What is JSON?

JSON (JavaScript Object Notation) is a light weight format used to store and exchange data. It is used because it is easy for humans to read, easy for machines to process thus commonly used in APIs.

Example!

```
{  "name" : "John",  
  "age" : 25,  
  "city" : "Johannesburg" }
```

23. What is Prompt Engineering?

It is the skill of writing effective instructions for AI systems. Prompt Engineering is important because better prompts produce more accurate and useful responses.

→ Poor prompt! "Tell me about SQL."

→ Improved prompt! "Explain SQL for a beginner data analytics student with simple examples."

→ Poor prompt! "Write code."

→ Improved prompt! "Write a Python program that calculates std student averages and explain each line."

24. What is a MAP Function?

A MAP function applies a function to every item in a collection.

Python example!

```
numbers = [1, 2, 3]
```

```
result = list (map (lambda x: x*2, numbers))
```

```
print (result)
```

Output! [2, 4, 6]

25 What is Cloud Computing

Cloud computing is the delivery of computing services over the internet.

Types:

- IaaS (Infrastructure as a Service)
 - Provides servers and storage. Eg. Amazon EC2.
- PaaS (Platform as a Service)
 - Provides a platform for developers. Eg. Google App Engine
- SaaS (Software as a Service)
 - Provides ready to use software online. Eg. Google Docs.

26. What is a Version Control System?

A Version control system tracks changes made to files and code over time. Git is a popular version control system used by developers and data professionals.

Git is used because it tracks project history, supports team work and allows rollback to previous versions.

Commit:

- A saved snapshot of changes.

Branch:

- A separate version of a project used for testing or new features.

27, What is a Jupyter Notebook?

It is an interactive environment used for writing and running code. It is commonly used with Python.

Why it is popular!

- Combines code, text and visualisations.
- Easy for data analysis.

- Supports experimentation and reporting.

28. What is Python?

Python is a high level programming language known for simplicity and readability. It is popular because it is beginner friendly, powerful for analytics and AI and ~~targeted~~ large community support.

Common Libraries:

1. Pandas - Data analysis and manipulation
2. NumPy - Numerical computing
3. Matplotlib - Data visualisation
4. Scikit-learn - Machine learning

Section D: Data Ethics, Governance & Careers

29. What is Data Privacy?

Data privacy is the protection of personal information from unauthorised access or misuse. It is important because personal data must be handled securely and responsibly.

GDPR (General Data Protection Regulation)!

- A data privacy law introduced by the European Union.
- Protects the personal data of EU citizens.
- Applies to organisations handling EU user data, even outside Europe.

30. What is Data Ethics?

Data ethics refers to the responsible and fair use of data and AI. Examples of unethical use:

1. Selling personal data without permission.
2. Using biased AI systems in hiring.
3. Spreading fake AI-generated content.

These are problematic because they violate privacy, fairness and trust.

31. What is Data Governance?

Data governance is the process of managing data quality, security, availability and policies. Its key components are; data quality, security and privacy, data standards, access control and compliance and regulations.

32. What is Bias in AI?

Bias in AI occurs when an AI system produces unfair or inaccurate results because of biased training data or design.

An example would be a facial recognition system performing poorly on certain ethnic groups. The consequences are discrimination, unfair treatment and loss of trust in AI systems.

33. Differences Between Data Roles.

Role	Main Responsibility	Common Skills	Common Tools
Data Analyst	Analyze and Visualize data	SQL, Excel, BI tools	Power BI, Tableau
Data Scientist	Build predictive models	Databases, ETL, cloud Python, ML, statistics	Sp Python, Scikit-learn
Data Engineer	Build data pipeline	Databases, ETL, Cloud	Spark, Air-Flow
AI Engineer	Build AI systems and applications	Deep learning, NLP	TensorFlow, Pytorch

34. What is Business Intelligence?

Business Intelligence (BI) refers to technologies and processes used to analyze data and support decision making.

Popular BI tools include:

- Power BI - Creates dashboards and reports.
- Tableau - Visualizes and analyzes data interactively.

These tools help businesses identify trends and make data-driven decisions.

Section E: Personal Reflection.

35. Most Interesting Concept

The most interesting concept was Artificial Intelligence and Large Language models because they can understand language, answer questions and help people solve problems in real time. It is surprising how AI can generate text, images and even code.

36. Most Difficult Concept

The most difficult concept was understanding deep learning and neural networks because they involve many layers and mathematical processes. Helpful resources included Youtube tutorials, ChatGPT explanations and beginner articles from Google.

37. How Data and AI will affect my Future.

Data and AI will affect my future career by creating opportunities in analytics, automation and decision making. In everyday life, AI is already used in navigation apps, streaming recommendations, online shopping and virtual assistants. In the future, data skills will help me make better business and career decisions.

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